MultiXplore: A context-dependent approach to capture insights in high-dimensional datasets

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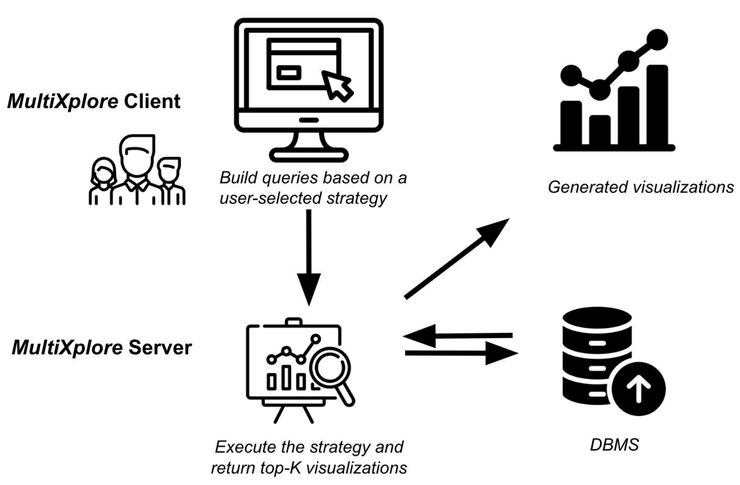
Introduction: Capturing insights from a high-dimensional dataset requires laborious explorations even for experienced data analysts. Previous methods like SeeDB[1], Zenvisage[2] and Foresight[3] have introduced different metrics to help expedite the initial exploration process. However, individuals may have different understandings towards insights, yet currently no method has provided a flexible framework to address such needs. Here we developed a context-dependent approach *MultiXplore* for users to navigate high-dimensional datasets by choosing different strategies based on their experiences.

Architecture:

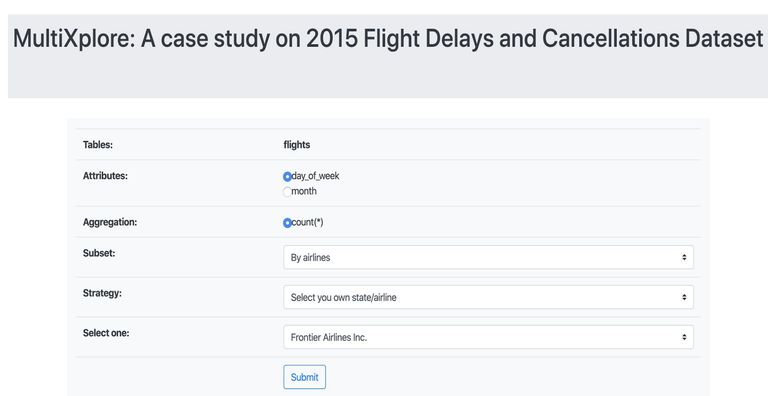


Methods: *MultiXplore* supports four different strategies:

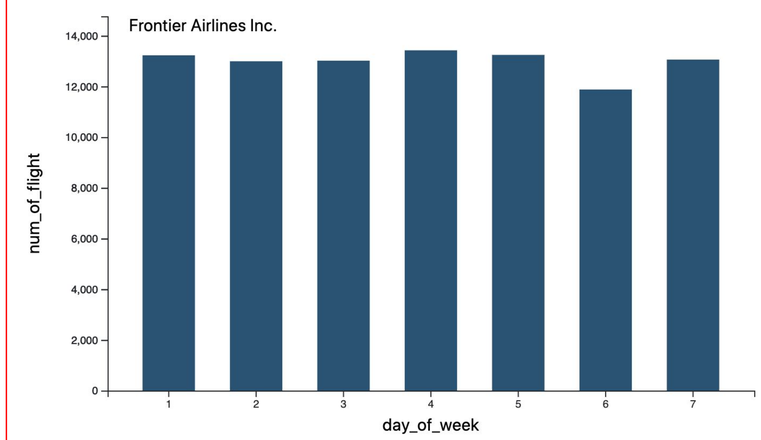
- **Divergence:** returns top-*K* plots of subsets that have the most different distributions (Kullback-Leibler divergence) than the plot on the aggregated data.
- **Statistical (**for advanced users): return top-*K* plots of subsets that have the highest values for a given metric, currently *MultiXplore* supports skewness, variance, tailedness, etc.
- **Random (**for inexperienced users or just for fun): randomly return a plot of a subset, and users can choose to get similar or dissimilar plots.
- **Select your own** (details on demand)**:** users can select a subset and plot it, and then get similar or dissimilar plots.

Evaluations: we use a flight delays/cancellations dataset of 2015 as a case study.

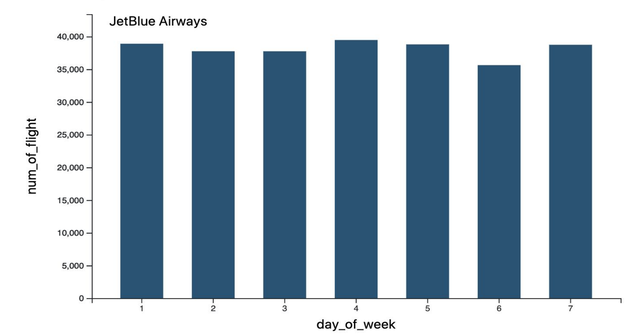
1. *User interface*

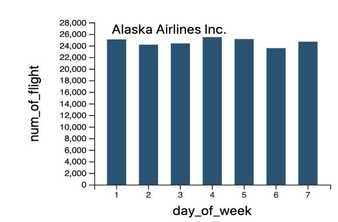


2. select your own: users select an airline:



3. Top-3 most similar plots returned by MultiXplore:





Conclusions: *MultiXplore* successfully integrates different popular recommendation strategies, and allows users to adopt them flexibly. We expect *MultiXplore* to help users to navigate more high-dimensional datasets in the future.

References: [1]Parameswaran et al, VLDB, 2013
[2]Siddiqui et al, VLDB, 2016
[3]Demiralp et al, *VLDB & KDD IDEA, 2017*